

Jeremiah Sol Larocco

Software Engineer | TypeScript, Node.js & Cross-Platform Applications

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PROFESSIONAL SUMMARY

Software engineer with 15+ years delivering production systems across geospatial, storage, and enterprise platforms. Expertise in TypeScript, Node.js, and event-driven architectures for applications with hardware protocols, embedded I/O, and timing-sensitive device interactions. Track record stabilizing cross-platform desktop software, modular API design, and CI/CD release quality. Background in defense-adjacent geospatial tooling, async workflow debugging, and test-driven development in remote teams.

WORK EXPERIENCE

Software Engineer

Dec, 2018 - Jun, 2021

Thermopylae Sciences + Technology

Onsite | Chantilly, Virginia

- Led stabilization and feature expansion of a **cross-platform geospatial desktop application** used for mission planning and situational awareness, improving reliability across Windows and macOS deployments serving defense and intelligence customers.
- Architected **event-driven, asynchronous workflows** connecting map visualization layers to live data feeds, resolving race conditions and timing-sensitive UI updates that previously caused intermittent rendering failures under heavy load.
- Designed modular **TypeScript and Node.js service layers** with clear dependency boundaries, enabling parallel team development and reducing merge conflicts across a fast-moving codebase.
- Built **REST and WebSocket APIs** for ingesting geospatial assets and telemetry streams, applying consistent validation patterns that cut integration defects reported during customer acceptance testing.
- Diagnosed complex **device and protocol interaction bugs** involving serial data parsers and external sensor feeds, using structured logging and replay tooling to isolate faults in multi-threaded I/O paths.
- Established **Jest unit and integration test suites** with interface mocks for hardware adapters, raising automated coverage on critical device-handling modules and catching regressions before release.
- Improved **CI/CD packaging and release pipelines** for desktop builds, standardizing artifact signing, version tagging, and smoke-test gates that shortened release validation cycles.
- Partnered with analysts and field engineers to translate **operational requirements into maintainable application features**, documenting architecture decisions that accelerated onboarding for new contributors.

Software engineer

Jul, 2013 - Nov, 2018

SolidFire, Inc.

Onsite | Boulder, Colorado

- Owned core modules of a **Node.js management platform** for all-flash storage arrays, delivering APIs and background services that orchestrated cluster provisioning across thousands of production nodes.
- Engineered **asynchronous, event-driven control flows** for volume lifecycle operations, eliminating deadlock scenarios in long-running tasks and improving recovery behavior after partial cluster failures.
- Refactored legacy JavaScript into a **modular TypeScript architecture** with explicit dependency injection, reducing circular imports and making feature modules independently testable.
- Designed **REST API contracts** consumed by web consoles and automation clients, applying versioning and pagination patterns that supported backward-compatible releases across multiple product lines.
- Debugged **timing-sensitive protocol interactions** between management services and storage controllers, tracing byte-level request/response sequences to resolve intermittent timeout errors in the field.
- Built comprehensive **Jest and integration test harnesses** with mocked device interfaces, enabling reliable simulation of hardware fault paths without physical lab dependencies.
- Strengthened **CI/CD build and release automation** with gated test stages, artifact promotion rules, and rollback procedures that improved deployment confidence for monthly product updates.

- Mentored engineers on **clean code practices, code review standards, and test design**, raising review throughput while maintaining low defect escape rates on customer-facing features.
- Collaborated with support and field teams to prioritize **production incident fixes**, delivering targeted patches that restored service levels without destabilizing adjacent platform components.

Software Engineer

Mar, 2009 - Jun, 2013

Spatial Corp

Onsite | Broomfield, Colorado

- Developed features for a **cross-platform 3D modeling SDK** used in CAD and engineering applications, extending geometry kernels and export pipelines consumed by desktop and embedded clients worldwide.
- Implemented **JavaScript and C++ interoperability layers** enabling plugin authors to script complex modeling workflows while preserving performance on compute-intensive operations.
- Resolved **asynchronous callback and threading defects** in native libraries called from managed UI threads, stabilizing integrations where timing-sensitive host callbacks caused intermittent crashes.
- Designed **modular plugin APIs** with versioned interfaces and dependency isolation, allowing third-party extensions to ship independently of core platform releases.
- Created **automated unit and regression tests** for geometry algorithms and file-format converters, catching precision and compatibility regressions before they reached OEM partners.
- Optimized **memory and I/O performance** for large assembly loads, reducing peak resource usage and improving responsiveness for customers working with multi-gigabyte model files.
- Partnered with application engineers across **aerospace and manufacturing accounts** to diagnose integration issues, translating field feedback into durable SDK improvements and clearer developer documentation.

Application Developer

Mar, 2006 - Feb, 2007

Verizon Business

Remote | New York

- Built and maintained **Java and JavaScript web applications** supporting enterprise customer portals, delivering form workflows and reporting views used by operations teams daily.
- Integrated **REST-style backend services** with relational data stores, implementing validation and error-handling patterns that reduced user-facing failures on high-traffic account pages.
- Participated in **Git-based code reviews and release cycles**, fixing defects identified in QA and contributing patches that improved stability ahead of scheduled deployments.
- Documented **API usage and deployment steps** for internal support teams, shortening handoff time when production issues required rapid developer triage.
- Collaborated remotely with **cross-functional stakeholders** to clarify requirements and deliver incremental enhancements within fixed sprint commitments.

EDUCATION

University of Colorado Colorado Springs

Bachelor of Science, Computer Science

SKILLS

Frontend: TypeScript, JavaScript, React, HTML5, CSS3, Electron, Cross-Platform Desktop UI, Event-Driven UI Patterns

Backend: Node.js, REST APIs, WebSockets, Asynchronous Workflows, Microservices, Python, C++, Java

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQL Query Optimization

Cloud & DevOps: Docker, AWS, Azure, CI/CD Pipelines, Build Automation, Git, Release Packaging

Monitoring & Tools: Jest, Unit & Integration Testing, Mocking Frameworks, Structured Logging, Serial/USB Device Debugging, GitHub

Workflow & PM: Agile/Scrum, Remote Collaboration, Technical Documentation, Code Review, Mentorship, Stakeholder Communication